

# Paarth Manchanda

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## EDUCATION

|                                   |           |                                           |        |
|-----------------------------------|-----------|-------------------------------------------|--------|
| B.TECH(Mathematics and Computing) | 2025-2029 | Delhi Technological University, New Delhi | 7.8    |
| CBSE (Class XII)                  | 2025      | P. D. Model School                        | 92.2 % |
| CBSE (Class X)                    | 2023      | St. Mary's Sr. Sec. School                | 91.6 % |

## PROJECTS

**DacSaab** — *Offline Multimodal AI Health Assistant* | Multimodal Architecture, LLMs, Edge AI

- Developed an offline-capable healthcare assistant for rural health workers using **quantized LLMs**, **XGBoost-based clinical triage models**, and multimodal pipelines supporting **text, voice, and medical image inputs**, achieving sub-2s local inference on low-resource devices.
- Built an edge-deployable Android/Desktop solution using **Flutter, ONNX Runtime, and PyTorch**, integrating local-language speech processing, symptom-to-disease prediction, and image-based condition analysis for **3+ multimodal input streams** and structured patient summarization workflows for treatment guidance.

**AI-Interview** — *Real-time mock interview & feedback platform* | Gen-AI, Speech Processing, Full-Stack Development

- Developed a conversational mock interview platform using a **fine-tuned Mistral 7B LLM with QLoRA**, enabling contextual interview simulations, grammar-aware feedback, and adaptive conversational workflows for voice-based interview practice.
- Integrated **Faster-Whisper and CTranslate2** for low-latency speech-to-text transcription, achieving **~1-2s response latency** during inference while supporting real-time voice/text interactions, speaking pace analysis, and Flask-based deployment pipelines optimized for conversational AI workloads.

**TerrainSeg** — *Cross-Domain Offroad Segmentation Pipeline* | Computer Vision, Semantic Segmentation

- Developed an off-road semantic segmentation pipeline during a competitive hackathon using **SegFormer-based architectures**, processing **1,000+ annotated terrain images** for obstacle-aware scene parsing and autonomous navigation in unstructured environments.
- Trained the model on one environmental distribution and evaluated robustness under significant **cross-domain distribution shifts**, implementing preprocessing, augmentation, and PyTorch-based inference optimization workflows to improve segmentation consistency across **multiple terrain and lighting conditions**.

## TECHNICAL SKILLS

- Languages:** Python, C/C++, Javascript, HTML/CSS, Ruby, Dart
- Frameworks and Libraries:** Pytorch, Tensorflow, Flask, OpenCV, TailwindCSS, ReactJS, ExpressJS
- Domains:** Machine Learning, Deep Learning, Computer Vision, NLP, Data Visualization
- Certifications:** [Coursera ML Specialization](#), [Coursera DL Specialization](#)